PROJECT : Villa Complex 415V LV Network	DETAILS OF CONNECTED LOAD, MAX DEMAND & kWh METERING			AREA : VILLA, UAE
PLANNED COMPLETION DATE :	OWNER : Mr. Ahmed Al Mansoori			PLOT No: 123-456
SUMMARY OF THE TCL AT TRANSFORMER LEVEL	CONSULTANT : ABC Engineering Consultants			LOC: MAIN BUILDING – GROUND FLOOR

TRANSFORMER REFERENCE	SP/TP	RATING - AMPS		FAULT DUTY kA	CABLE SIZE			ECC SIZE 1C, mm2	CONNECTION LOAD - kW			TCL (kW)	D.F	MDL (kW)	kWH METER			Remarks
		ACB	MCCB		PVC/XLPE/SWA/ PVC	2/4X1C MM2	2/3/4C MM2		R-PHASE kW	Y-PHASE kW	B-PHASE kW				1 - 0 (1)	3 - 0 (2)	CT (3)	
SUBSTATION-01																		
MDB-1	4P	1600 @ 0.90		36	BY DEWA				171.69	171.27	171.02	513.97	0.8	411.18			1	1500/5A CT
		TOTAL CONNECTED - LOAD PER PHASE							**171.69**	**171.27**	**171.02**	**513.97**		**411.18**			**1**	1X 1500/5A CT
		DIVERSITY FACTOR			**0.8**	**TOTAL**												
		MAX. DEMAND - 3 PHASE			**411.2**	kW		TCL =	**514**	kW								
								TCL (DUTY)=	**514**	kW								

PROJECT:	**Villa Complex 415V LV Network**				**DETAILS OF CONNECTED LOAD, MAXIMUM DEMAND & kWh METERING**													AREA : VILLA, UAE
PLANNED COMPLETION DATE:					OWNER: Mr. Ahmed Al Mansoori													PLOT NO: 123-456
MDB-1 (FED FROM DEWA)					CONSULTANT: ABC Engineering Consultants													LOCATION : Main building – ground floor
CIRCUIT/FEEDER SMDB/ DB NO.	SP/ TP	RATING (AMPS)			FAULT DUTY kA	CABLE SIZE, TYPE & No.OF CORES			ECC SIZE 1C mm2	LENGTH OF CABLE (Mtrs)	CONNECTED LOAD kW			TOTAL CONNECTED /INSTALLED LOAD (TCL) Kw	MAXIMUM DEMAND/ OPERATIONAL LOAD (MDL) Kw	PROPOSED TYPE & No OF kWh METER			REMARKS
		ACB	MCCB	ISOL		NO. OF CORES: 1C/2C/4C	TYPE: XLPE/PVC/SWA	SIZE			R-PH kW	Y-PH kW	B-PH kW			1-PH (1)	3-PH (2)	LV/HV-CT (3)	
INCOMER	TP		1600			BY DEWA												1	
OUT GOING																			
SMDB-GF (GROUND FLOOR SMDB)	TP		500		36	4C	XLPE/PVC/SWA	300	150	35	77.35	76.93	76.68	230.97	203.25				
SMDB-FF (FIRST FLOOR SMDB)	TP		200		36	4C	XLPE/PVC/SWA	95	50	55	36.33	36.33	36.33	109.00	95.92				
MCC-1 (PUMP ROOM MCC)	TP		315		36	4C	XLPE/PVC/SWA	150	95	40	43.33	43.33	43.33	130.00	114.40				
EV CHARGING – PARKING	TP		63		36	4C	XLPE/PVC/SWA	10	10	60	14.67	14.67	14.67	44.00	38.72				
SOLAR PV INVERTER	TP		100		36	4C	XLPE/PVC/SWA	25	16	30	0.00	0.00	0.00	0.00	0.00				
MDB CONNECTED TO: DEWA METER CABINET DB's CONNECTED TO: MDB-1						TOTAL CONNECTED LOAD PER PHASE					**171.69**	**171.27**	**171.02**	**513.97**	**452.29**			**1**	**TOTAL**
DEMAND FACTOR : 0.88					**MAX. DEMAND: 452.29 KW**						**TOTAL CONNECTED LOAD : 513.97 KW**							TOTAL BUILD UP AREA (Sq. mtr) :	
CONSULTANT/ CONTRACTOR:					TEL:									FAX :					
Type of Meter (Rating of incomer) : (1) Up to 60A (1 Phase) (2) Up to 125A (3 Phase) (3) LV CT / HV CT																			

PROJECT:	Villa Complex 415V LV Network				DETAILS OF CONNECTED LOAD, MAXIMUM DEMAND & kWh METERING													AREA : VILLA, UAE			
PLANNED COMPLETION DATE:											OWNER: Mr. Ahmed Al Mansoori							PLOT NO: 123-456			
SMDB-GF (FED FROM MDB-1)											CONSULTANT: ABC Engineering Consultants							LOCATION : Ground floor electrical room			
CIRCUIT/FEEDER SMDB/ DB NO.	SP/ TP	RATING (AMPS)			FAULT DUTY kA	CABLE SIZE, TYPE & No.OF CORES			ECC SIZE 1C mm2	LENGTH OF CABLE (Mtrs)	CONNECTED LOAD kW			TOTAL CONNECTED /INSTALLED LOAD (TCL) Kw	MAXIMUM DEMAND/ OPERATIONAL LOAD (MDL) Kw	PROPOSED TYPE & No OF kWh METER			REMARKS		
		ACB	MCCB	ISOL		NO. OF CORES: 1C/2C/4C	TYPE: XLPE/PVC/SWA	SIZE			R-PH kW	Y-PH kW	B-PH kW			1-PH (1)	3-PH (2)	LV/HV-CT (3)			
INCOMER	TP		500		36	4C	XLPE/PVC/SWA	300	150	35											
OUT GOING																					
LIGHTING	TP		80		25	4C	XLPE/PVC/SWA	16	16	30	15.00	15.00	15.00	45.00	39.60						
SOCKETS	TP		40		25	4C	XLPE/PVC/SWA	4	4	35	9.33	9.33	9.33	28.00	24.64						
HVAC	TP		250		25	4C	XLPE/PVC/SWA	95	50	40	40.00	40.00	40.00	120.00	105.60						
DB-GF1 (VILLA GROUND FLOOR DB)	TP		100		25	4C	XLPE/PVC/SWA	35	16	30	13.02	12.60	12.35	37.97	33.41						
SMDB CONNECTED TO: MDB-1 DB's CONNECTED TO: SMDB-GF						TOTAL CONNECTED LOAD PER PHASE					**77.35**	**76.93**	**76.68**	**230.97**	**203.25**				**TOTAL**		
DEMAND FACTOR : 0.88					**MAX. DEMAND: 203.25 KW**						**TOTAL CONNECTED LOAD : 230.97 KW**							TOTAL BUILD UP AREA (Sq. mtr) :			
CONSULTANT/ CONTRACTOR:							TEL:								FAX :						
Type of Meter (Rating of incomer) : (1) Up to 60A (1 Phase) (2) Up to 125A (3 Phase) (3) LV CT / HV CT																					

PROJECT:	**Villa Complex 415V LV Network**				**DETAILS OF CONNECTED LOAD, MAXIMUM DEMAND & kWh METERING**											AREA : VILLA, UAE			
PLANNED COMPLETION DATE:					OWNER: Mr. Ahmed Al Mansoori											PLOT NO: 123-456			
SMDB-FF (FED FROM MDB-1)					CONSULTANT: ABC Engineering Consultants											LOCATION : First floor electrical room			
CIRCUIT/FEEDER SMDB/ DB NO.	SP/ TP	RATING (AMPS)			FAULT DUTY kA	CABLE SIZE, TYPE & No.OF CORES			ECC SIZE 1C mm2	LENGTH OF CABLE (Mtrs)	CONNECTED LOAD kW			TOTAL CONNECTED /INSTALLED LOAD (TCL) Kw	MAXIMUM DEMAND/ OPERATIONAL LOAD (MDL) Kw	PROPOSED TYPE & No OF kWh METER			REMARKS
		ACB	MCCB	ISOL		NO. OF CORES: 1C/2C/4C	TYPE: XLPE/PVC/SWA	SIZE			R-PH kW	Y-PH kW	B-PH kW			1-PH (1)	3-PH (2)	LV/HV-CT (3)	
INCOMER	TP		200		36	4C	XLPE/PVC/SWA	95	50	55									
OUT GOING																			
OFFICE	TP		63		25	4C	XLPE/PVC/SWA	10	10	25	11.67	11.67	11.67	35.00	28.00				
IT ROOM	TP		40		25	4C	XLPE/PVC/SWA	4	4	20	6.67	6.67	6.67	20.00	16.00				
LAB	TP		80		25	4C	XLPE/PVC/SWA	16	16	30	16.67	16.67	16.67	50.00	40.00				
CORRIDOR LIGHTING	SP		25		25	2C	XLPE/PVC/SWA	6	6	35	1.33	1.33	1.33	4.00	3.20				
SMDB CONNECTED TO: MDB-1						TOTAL CONNECTED LOAD PER PHASE					**36.33**	**36.33**	**36.33**	**109.00**	**87.20**				**TOTAL**
DEMAND FACTOR : 0.80					**MAX. DEMAND: 87.20 KW**						**TOTAL CONNECTED LOAD : 109.00 KW**					TOTAL BUILD UP AREA (Sq. mtr) :			
CONSULTANT/ CONTRACTOR:					TEL:									FAX :					
Type of Meter (Rating of incomer) : (1) Up to 60A (1 Phase) (2) Up to 125A (3 Phase) (3) LV CT / HV CT																			

PROJECT: Villa Complex 415V LV Network

LOCATION OF DB: Villa GF – store room

DB NO : DB-GF1

FED FROM : SMDB-GF

LOAD DISTRIBUTION SCHEDULE

(3 PHASE)

RATING OF INCOMER	RATING OF ELCB	SL NO.	CIR NO.	MCB RAT. IN AMPS	CCT WIRE SIZE (mm2)	ECC WIRE SIZE (mm2)	ROOM/ AREA	CONNECTED LOAD/ POINTS															WATTS/ UNIT	LOAD/CIRCUIT (WATTS)			REMARKS
								LTG	EX.FAN	SH.S/O	13A S.S/O	13A T.S/O	W/H	FCU	SPUR O/L	ISOL	COOKER	S A/C	WAT. PUMP	OTH	C.FAN	15A S/O		R	Y	B	
100A TP ISOLATOR	ELCB-1 25 A 100 mA 4P	1	R1	10	2.5	2.5	Living lighting	12															100	1200			
		2	Y1	6	2.5	2.5	Majlis lighting	10															100		1000		
		3	B1	6	2.5	2.5	External lighting	10															100			1000	
			R2																								
		4	Y2	6	2.5	2.5	Kitchen lighting	8	1														100 / 40		840		
		5	B2	6	2.5	2.5	Dining lighting	6															100			600	
	ELCB-2 40 A 30 mA 4P	6	R3	40	6	6	Cooker										1						6000	6000			
		7	Y3	20	4	4	Guest bath water heater						1										3000		3000		
		8	B3	20	4	4	Kitchen water heater						1										3000			3000	
		9	R4	16	4	4	Guest bedroom split A/C											1					2500	2500			
		10	Y4	16	4	4	Living split A/C											1					2500		2500		
		11	B4	16	4	4	Majlis split A/C											1					2500			2500	
	ELCB-3 25 A 30 mA 4P	12	R5	10	4	4	Guest bedroom	4			4										1		100 / 250 / 80	1480			
		13	Y5	16	4	4	Living sockets				8												250		2000		
		14	B5	16	4	4	Dining split A/C											1					2500			2500	
		15	R6	6	4	4	Maid room	3	1		3												100 / 40 / 250	1090			
		16	Y6	16	4	4	Laundry															2	1000		2000		
		17	B6	10	4	4	Majlis sockets				6												250			1500	
	ELCB-4 25 A 30 mA 4P	18	R7	6	4	4	Water pump												1				750	750			
		19	Y7	6	4	4	Dining sockets				4												250		1000		
		20	B7	10	4	4	Kitchen sockets				5												250			1250	
			R8																								
		21	Y8	6	4	4	Guest bathroom	2	1	1													100 / 40 / 20		260		
			B8																								
			R9																								
			Y9																								
			B9																								
			R10																								
			Y10																								
			B10																								
																							TOTAL (kW)	**13.02**	**12.60**	**12.35**	**37.97**

CABLE SIZE: 4C 35mm² Cu XLPE/SWA/PVC + 1C 16mm² Cu/PVC G/Y AS ECC, 30 m

Columns: LTG, EX.FAN, SH.S/O, 13A S.S/O, 13A T.S/O, W/H, FCU, SPUR O/L, ISOL, COOKER, S A/C, WAT.PUMP, OTH, C.FAN, 15A S/O. C. FAN = Ceiling fan, EX. FAN = Exhaust fan, W/H = Water heater, SH. S/O = Shaver socket-outlet, S = Split type

PROJECT:	**Villa Complex 415V LV Network**	**DETAILS OF CONNECTED LOAD, MAXIMUM DEMAND & kWh METERING**	AREA : VILLA, UAE
PLANNED COMPLETION DATE:		OWNER: Mr. Ahmed Al Mansoori	PLOT NO: 123-456
MCC-1 (FED FROM MDB-1)		CONSULTANT: ABC Engineering Consultants	LOCATION : Basement pump room

CIRCUIT/FEEDER SMDB/ DB NO.	SP/ TP	RATING (AMPS)			FAULT DUTY kA	CABLE SIZE, TYPE & No.OF CORES			ECC SIZE 1C mm2	LENGTH OF CABLE (Mtrs)	CONNECTED LOAD kW			TOTAL CONNECTED /INSTALLED LOAD (TCL) Kw	MAXIMUM DEMAND/ OPERATIONAL LOAD (MDL) Kw	PROPOSED TYPE & No OF kWh METER			REMARKS
		ACB	MCCB	ISOL		NO. OF CORES: 1C/2C/4C	TYPE: XLPE/PVC/SWA	SIZE			R-PH kW	Y-PH kW	B-PH kW			1-PH (1)	3-PH (2)	LV/HV-CT (3)	
INCOMER	TP		315		36	4C	XLPE/PVC/SWA	150	95	40									
OUT GOING																			
WATER PUMP	TP		160		25	4C	XLPE/PVC/SWA	50	25	20	25.00	25.00	25.00	75.00	75.00				
FIRE PUMP	TP		125		25	4C	XLPE/PVC/SWA	35	16	25	18.33	18.33	18.33	55.00	55.00				
MCC CONNECTED TO: MDB-1						TOTAL CONNECTED LOAD PER PHASE					**43.33**	**43.33**	**43.33**	**130.00**	**130.00**				**TOTAL**

DEMAND FACTOR : 1.00 **MAX. DEMAND: 130.00 KW** **TOTAL CONNECTED LOAD : 130.00 KW** TOTAL BUILD UP AREA (Sq. mtr) :

CONSULTANT/ CONTRACTOR: TEL: FAX :

Type of Meter (Rating of incomer) : (1) Up to 60A (1 Phase) (2) Up to 125A (3 Phase) (3) LV CT / HV CT